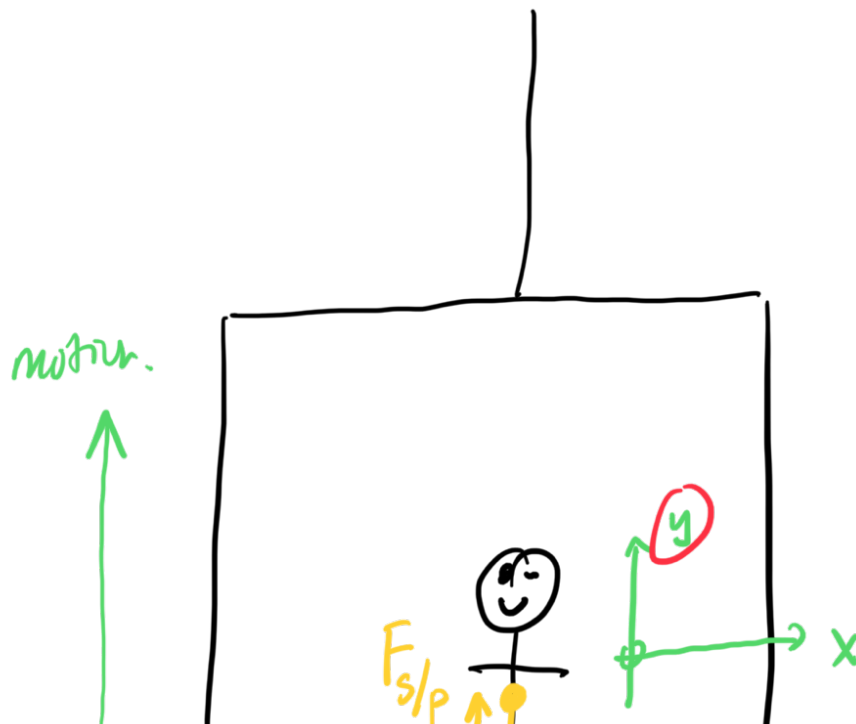


Physics 201 - Lecture 15

- finish assignment at 4
 - more on friction.
 - start assignment 5
(We will go slowly through each problem!)
-

9.





→ Free Body Diagram(s) for the objects of interest.

- ① gravity
- ② contact forces || Person.

Man ... $\Sigma F_y = m a_y$

$\textcircled{F_{s/p}} - \overset{\checkmark}{m} \overset{\checkmark}{g} = m \overset{\checkmark}{a}_y$

a) $a_y = 1.25 \text{ m/s}^2$ (+ve because "upwards")

$$F_{s/p} = m a_y + m g$$

$$= (88)(1.25) + (88)(9.8)$$

$$= 972 \text{ N}$$

$$\vec{F}_{S/P} = 972 \text{ N upwards.}$$

$$\vec{F}_{P/S} = 972 \text{ N downwards.}$$

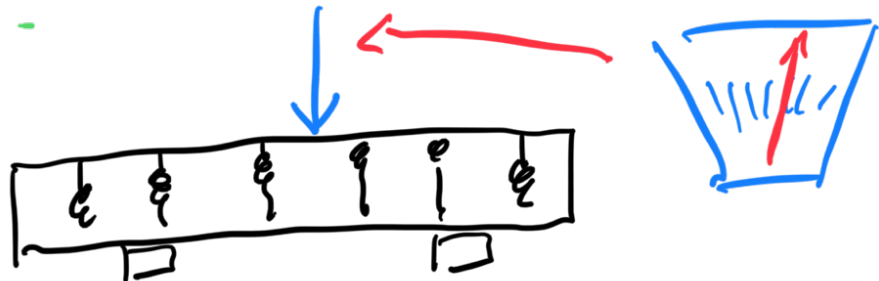
$$\vec{F}_{S/P} = -\vec{F}_{P/S}$$

Newton's 3rd Law

Scale



How do scales work?



" In a velocity $\rightarrow a = 0$

b)

constant velocity

$$F_{s/p} - mg = ma_y$$

$$F_{s/p} = mg = 862 \text{ N}$$

$$\vec{F}_{s/p} = 862 \text{ N upwals.}$$

$$\vec{F}_{p/s} = \boxed{826 \text{ N}} \text{ downwals.}$$

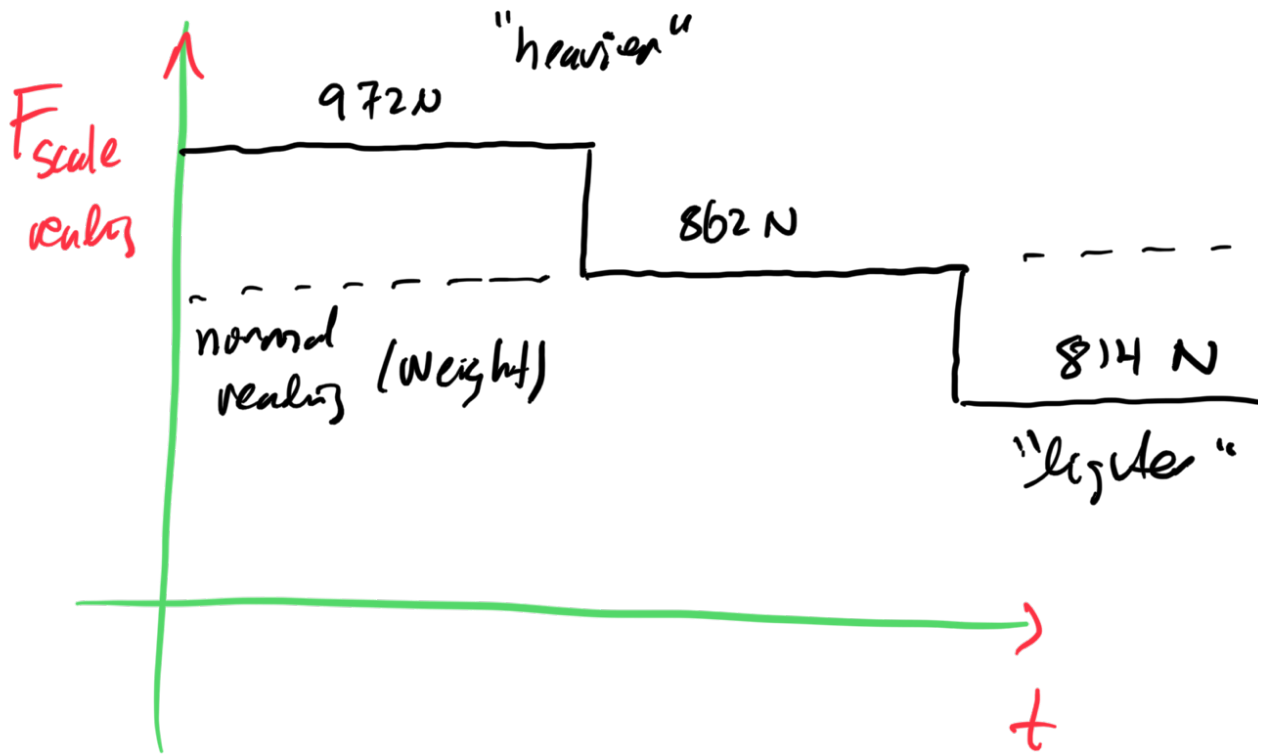
c) $a_y = -0.550 \text{ m/s}^2$

$$F_{s/p} - mg = ma_y$$

$$F_{s/p} = ma_y + mg$$

$$= (88)(-0.55) + (88)(9.8)$$

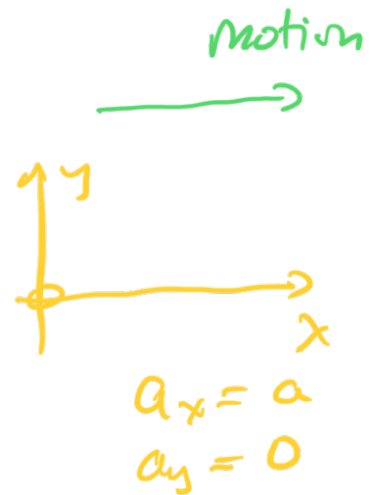
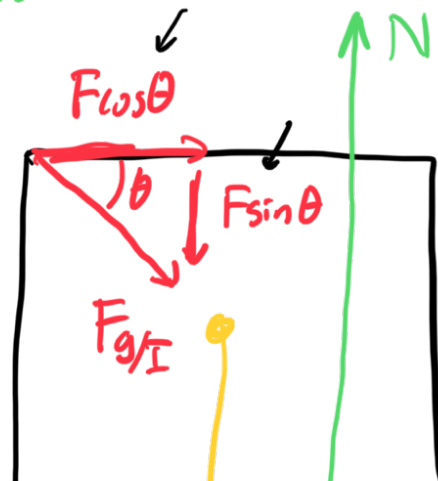
$$= 814 \text{ N}$$

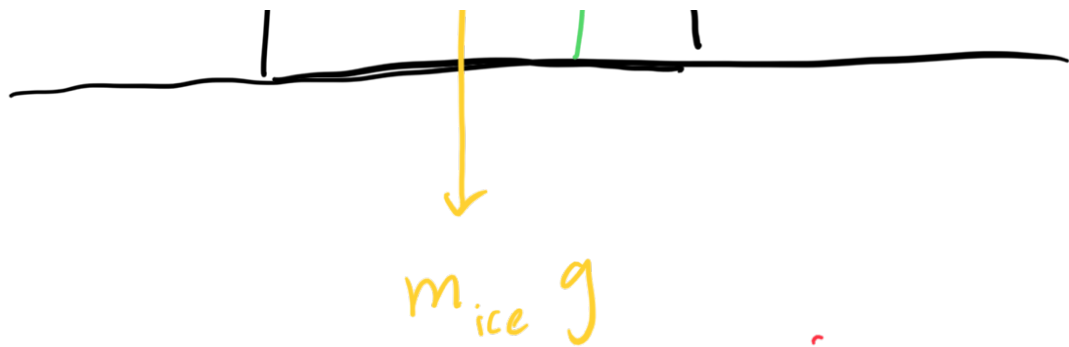


10. Object of interest ICE

"friction is small"

10
10





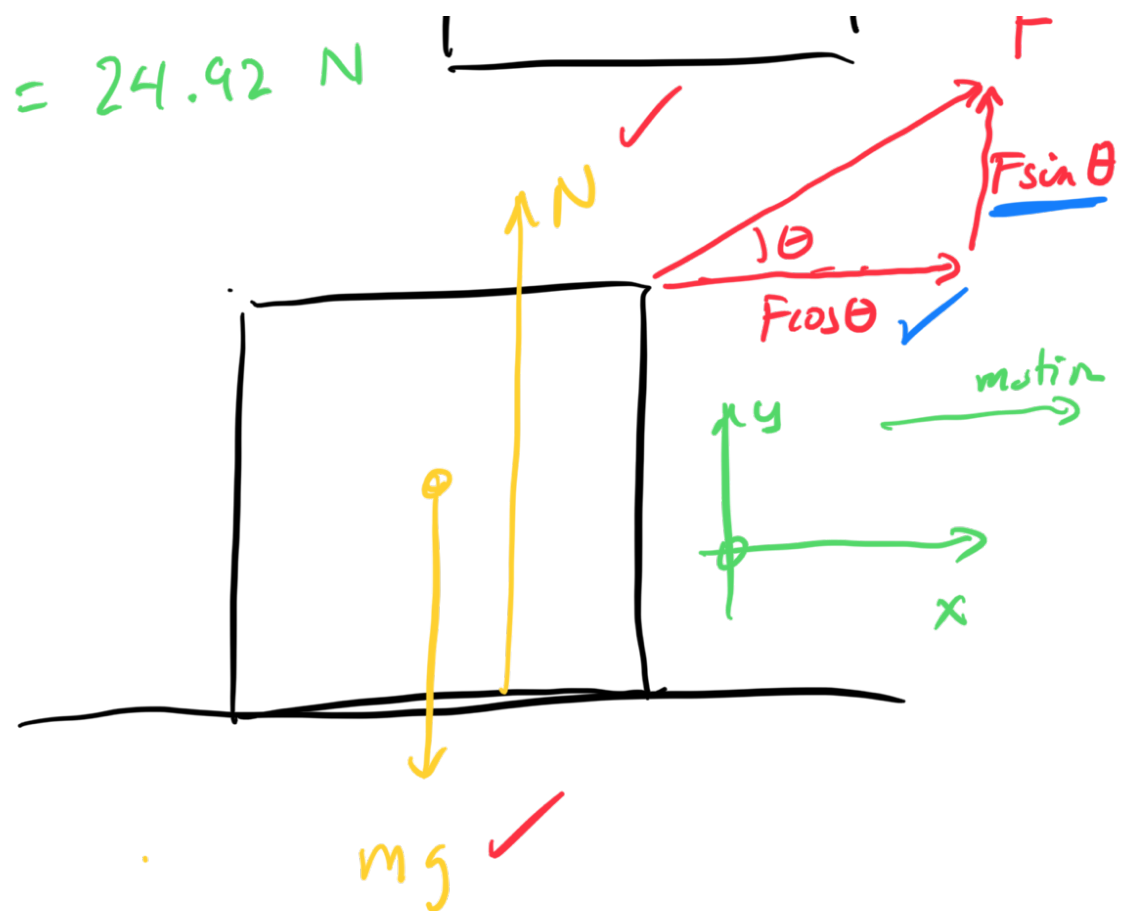
- ① identify all forces
- ② coordinate system
- ③ resolve into components.

X	Y
$a_x = a$	$a_y = 0$
$\Sigma F_x = ma_x$	$\Sigma F_y = ma_y = 0$
$F \cos \theta = ma_x$	$N - mg - F \sin \theta = 0$

$$\begin{aligned}
 F &= \frac{ma_x}{\cos \theta} \\
 &= \frac{(40)(1.550)}{\cos(28^\circ)}
 \end{aligned}$$

$$\begin{aligned}
 N &= mg + F \sin \theta \\
 &= (40)(9.8) + (24.92) \sin(28^\circ)
 \end{aligned}$$

$N = 404 \text{ N}$

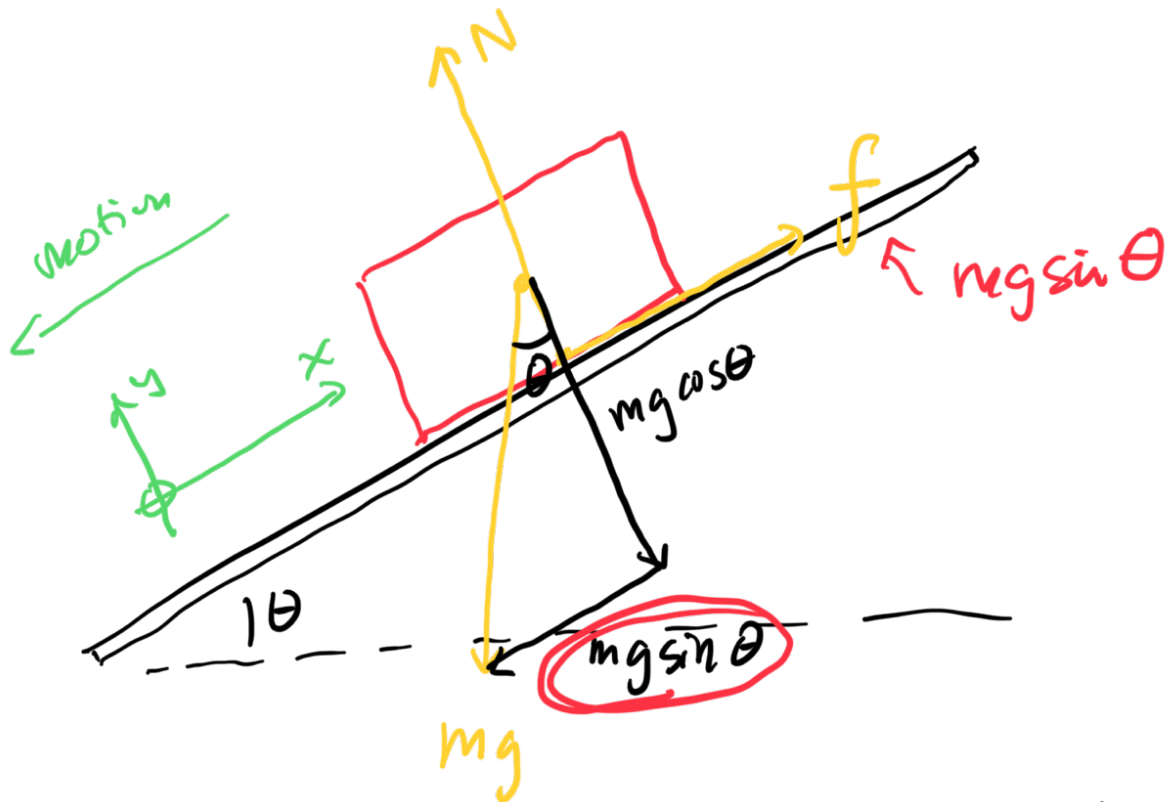


X	Y
$F \cos \theta = m a_x$ $\uparrow$ $F = 24.92 \text{ N}$	$N - mg + F \sin \theta = 0$ $N = mg - F \sin \theta$ $= (40)(9.8) - (24.92) \sin 28$ $= 380 \text{ N}$

Friction

→ Energy losses

→ \$\$\$



x	y
$\Sigma F_x = m a_x$	$\Sigma F_y = m a_y = 0$
$f - mg \sin \theta = m a_x$	$N - mg \cos \theta = 0$
	$N = mg \cos \theta$

Definition of
friction:

$$f = \mu N$$

coefficient of
friction.

$$f = \mu mg \cos \theta$$

$$\cancel{\mu mg \cos \theta} - \cancel{mg \sin \theta} = \cancel{m} a_x$$

$$a_x = \mu g \cos \theta - g \sin \theta$$

$$\Rightarrow a_x = g (\mu \cos \theta - \sin \theta)$$

TWO REGIMES:

Static

Kinetic

- object is not moving
relative to the surface.

- object is static
relative to the
surface.

$$a_x = 0 \quad \leftarrow \text{not moving}$$

$$0 = g (\underbrace{\mu \cos \theta - \sin \theta}_{=0})$$

$$\mu \cos \theta - \sin \theta = 0$$

$$\mu \cos \theta = \sin \theta$$

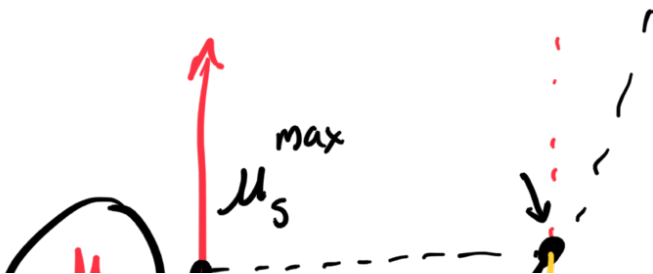
$$\mu = \frac{\sin \theta}{\cos \theta}$$

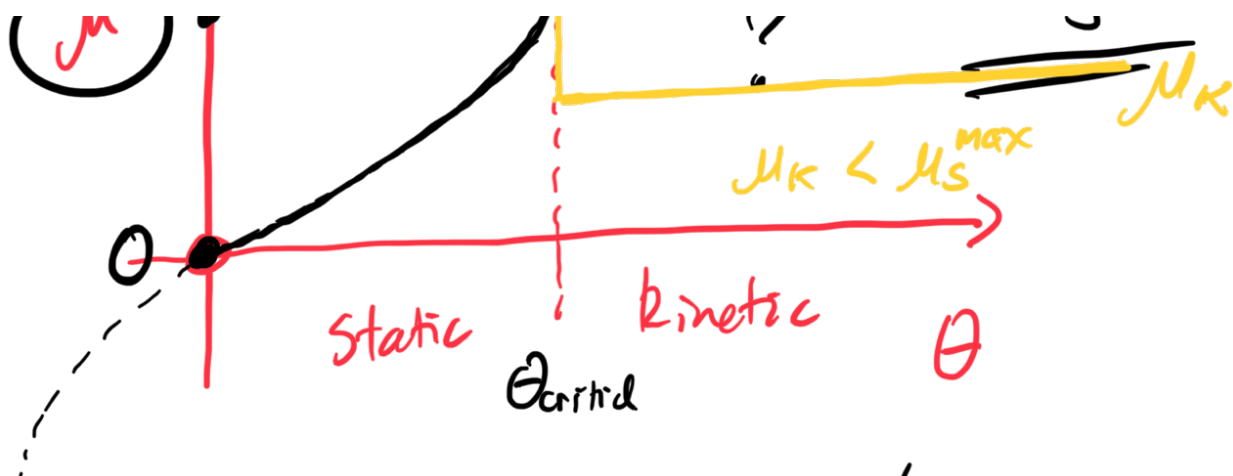
$$\boxed{\mu = \tan \theta} \quad \text{😊}$$

$$\mu_s^{\max} = \tan(\theta_{\text{critical}})$$

$$\theta_{\text{critical}} = \tan^{-1}(\mu_s)$$

$$\begin{aligned} f &= \mu mg \cos \theta \\ &= \tan \theta \cdot mg \cos \theta \\ &= \frac{\sin \theta}{\cos \theta} \cdot mg \cos \theta \\ &= mg \sin \theta \end{aligned}$$





← measured by experiment.
 $\mu_s^{max} \equiv$ max. coefficient of static friction.

↑ surface just gets overwhelmed (i)

Text books are sloppy!

" the ^{maximum} coefficient of static friction is 0.6 "

$$0 \leq \mu_s \leq \mu_s^{max}$$



Kinetic Friction (Sliding)

Coefficient of kinetic friction

$$= \mu_k \leftarrow \text{constant!}$$

(given)

Given: μ_s^{max} , μ_k

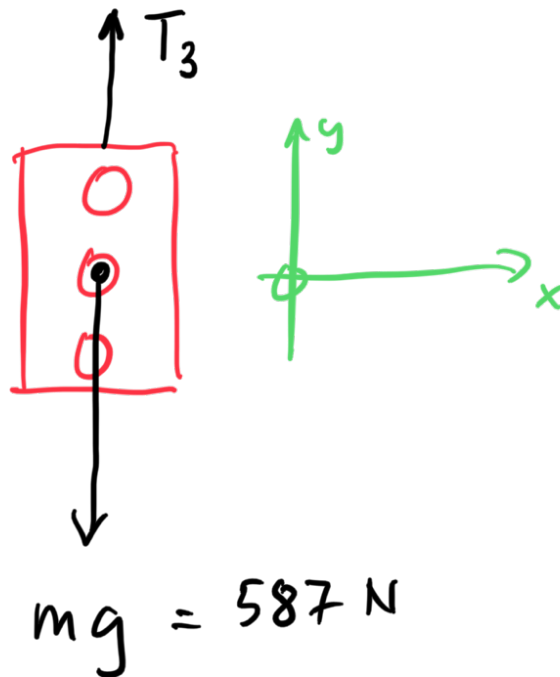
$$= 0.6 \quad = 0.4$$

Q1. What kind of problem is this?

"... no tension..."

Find the Force

→ Force problem → FBD



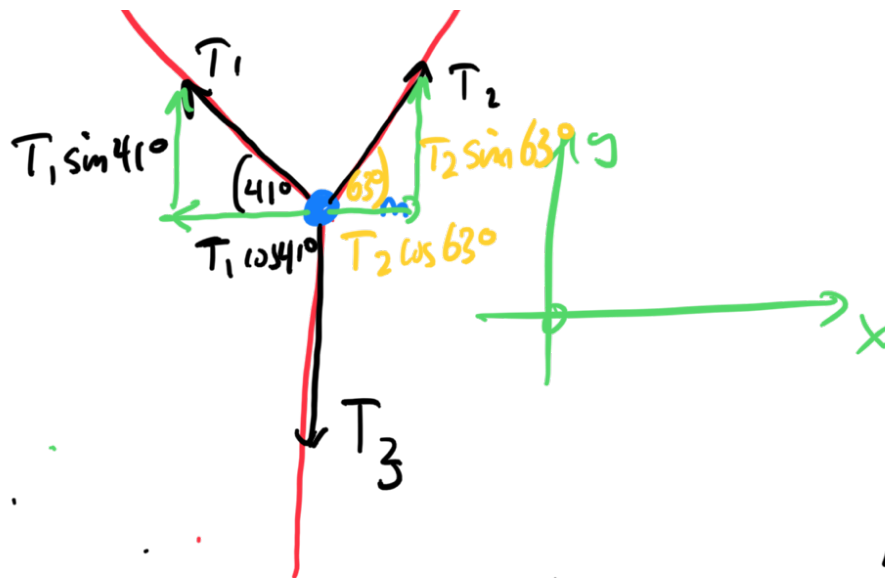
$$T_3 - mg = ma_y = 0$$

$$\therefore T_3 = mg = 587 \text{ N}$$

What is the object of interest?

~~41°~~

~~63°~~



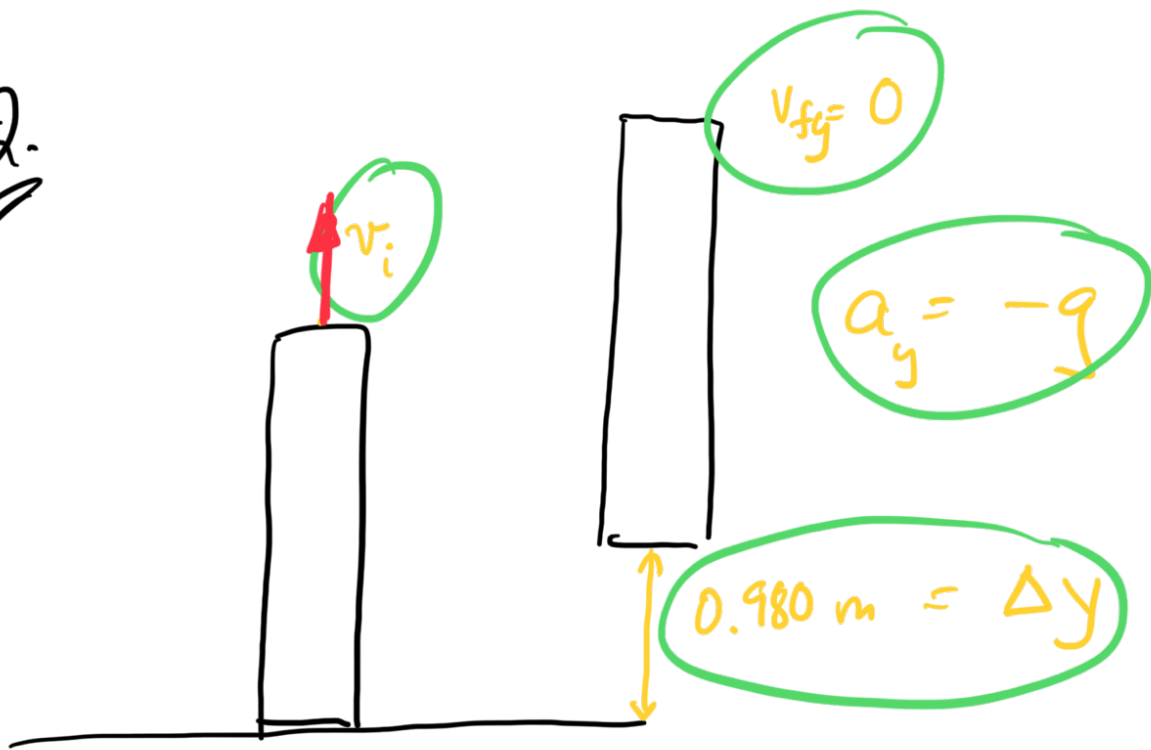
$$\begin{aligned}
 y: & \quad T_1 \sin 41^\circ + T_2 \sin 63^\circ - \cancel{T_3} = 0 \\
 x: & \quad -T_1 \cos 41^\circ + T_2 \cos 63^\circ = 0
 \end{aligned}$$

}
 algebra
 }
 trig.
 }
 solution

31% chance that I get it wrong.

$$T_1 = 275 \text{ N}, \quad T_2 = 457 \text{ N}$$

2.



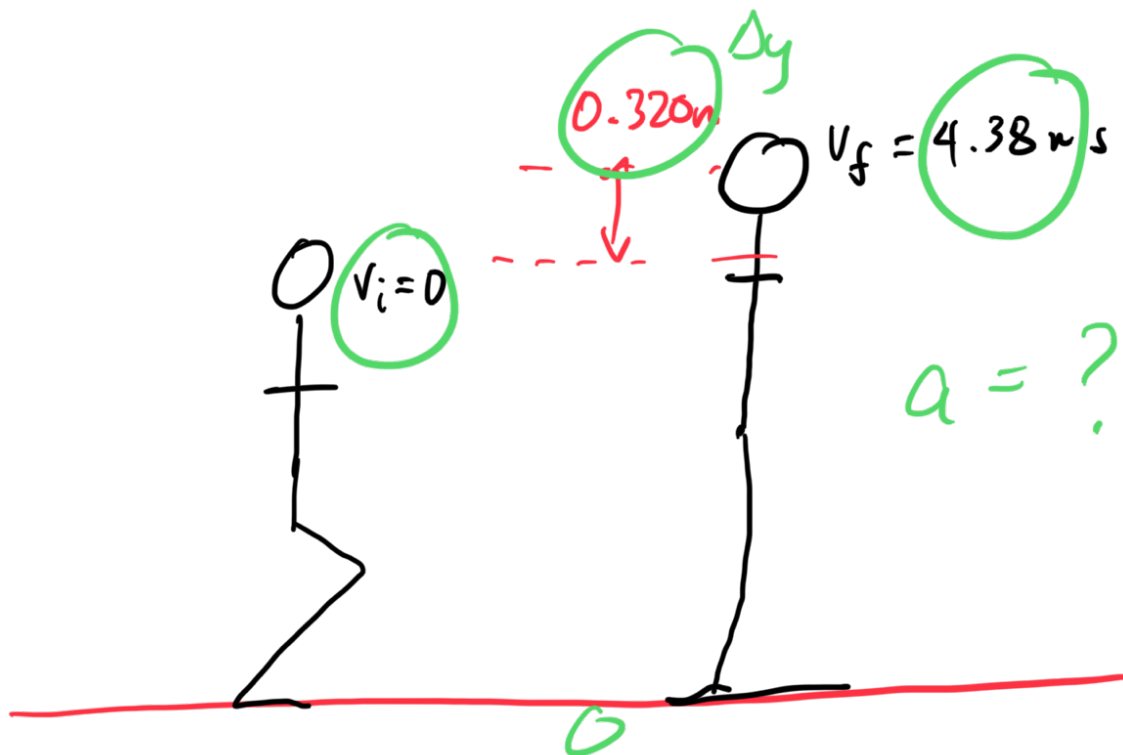
$$v_f^2 = v_i^2 + 2a_y \Delta y$$

$$0 = v_i^2 - 2g \Delta y$$

$$v_i^2 = 2g \Delta y$$

$$v_i = \pm \sqrt{2g \Delta y}$$

$$v_i = 4.38 \text{ m/s}$$



$$v_f^2 = \cancel{v_i^2} + 2a \Delta y$$

$$(4.38)^2 = 2(a)(.320)$$

$$a = \frac{(4.38)^2}{2(.320)} = 29.98 \text{ m/s}^2$$

$$m = 104 \text{ kg}$$

c) ~~"force"~~ $\rightarrow \vec{F} = ma$
 ~~104×29.98~~

~~$$= 3117.5 \text{ N}$$

$$3118$$~~

$$\vec{F}_{\text{net}} = m \vec{a}$$



$$-F_{\text{net}}^y = m a_y$$

$$F_{\text{floor/guy}} - mg = 3117.5 \text{ N}$$





$$\begin{aligned} \underline{F_{\text{floor/gung}}} &= 3117.5 + mg \\ &= 4140 \text{ N} \end{aligned} \quad \uparrow$$

$$F_{\text{gun/fo}} = 4140 \text{ N} \quad \downarrow$$